

Getting Setup on Windows

This document briefly describes how to get your Microsoft Windows (10) computer set up to create your own Gameboy Advance (GBA) programs using devkitPro.

- **devkitPro** is a toolchain (compiler, linker, libraries and other tools) that takes the source code and builds a working "ROM file" that can then be run in an emulator or on real hardware.
- **VisualBoyAdvance** is GBA emulator that allows you to run your newly created ROM file on your computer.

devkitPro

This is the development toolchain that is used to compile your source code into an executable binary. Without this package installed, you will not be able to build complete GBA/NDS programs (although you can still draw sprites and backgrounds).

Homepage: <http://www.devkitpro.org/>

For Windows, there is a graphical installer: <https://github.com/devkitPro/installer/releases/tag/v3.0.3>

Just use default settings in the installer.

When the installation is complete, verify that you have a bunch of new files and directories installed in your **c:\devkitpro** directory.

Now you should verify that your environment variables were set up properly (otherwise you won't be able to build). If the installation worked properly, then you will only need to verify that these variables were created. But there's a chance they weren't created (depending on your system's security settings), so it's a good idea to double-check.

- Open the "Start" menu and search "View Advanced System Settings"
- A Dialog box will appear. Near the bottom of this box will be an "Environment Variables" button. Click it.
- You should see the following variables/values or something similar (either in the top "User variables" or bottom "System variables" box):

```
DEVKITARM = /opt/devkitPro/devkitARM  
DEVKITPRO = /opt/devkitPro
```

- If you don't see those variables, then add them to the "User variables" (at the top) by clicking the "New" button and adding them one at a time. Note that there are a few other DEVKIT variables as well, but the 2 listed above are the only ones that matter.
- Finally, make sure that your "Path" environment variable contains **c:\devkitPro\msys2\bin**.
 - If you don't have a "Path" variable, then add a new one to the "User variables" that contains this string.
 - If you already have a "Path" variable, click on it, then click on the Edit button. Double click one of the empty rows in the dialog box that appears, then add the following string
c:\devkitPro\msys2\bin

VisualBoy Advance

VBA is the most popular GBA emulator. An emulator is a program that runs on a computer and pretends to be another device. This will allow us to test out our GBA programs on our development computer.

Homepage: <https://github.com/visualboyadvance-m/visualboyadvance-m/releases>

Installing VisualBoy Advance:

- Create a **c : \gamedev** folder on your drive.
- Go to the download page listed above and download the " visualboyadvance-m-Win-64bit.zip" file to your computer. Save the file in the **c : \gamedev** folder you created.
- Extract the contents of the zip file by right-clicking on the file and selecting "Extract all..."
- You will now have a folder in your **c : \gamedev** folder.

You can run VBA by double-clicking the .exe file in the extracted folder.